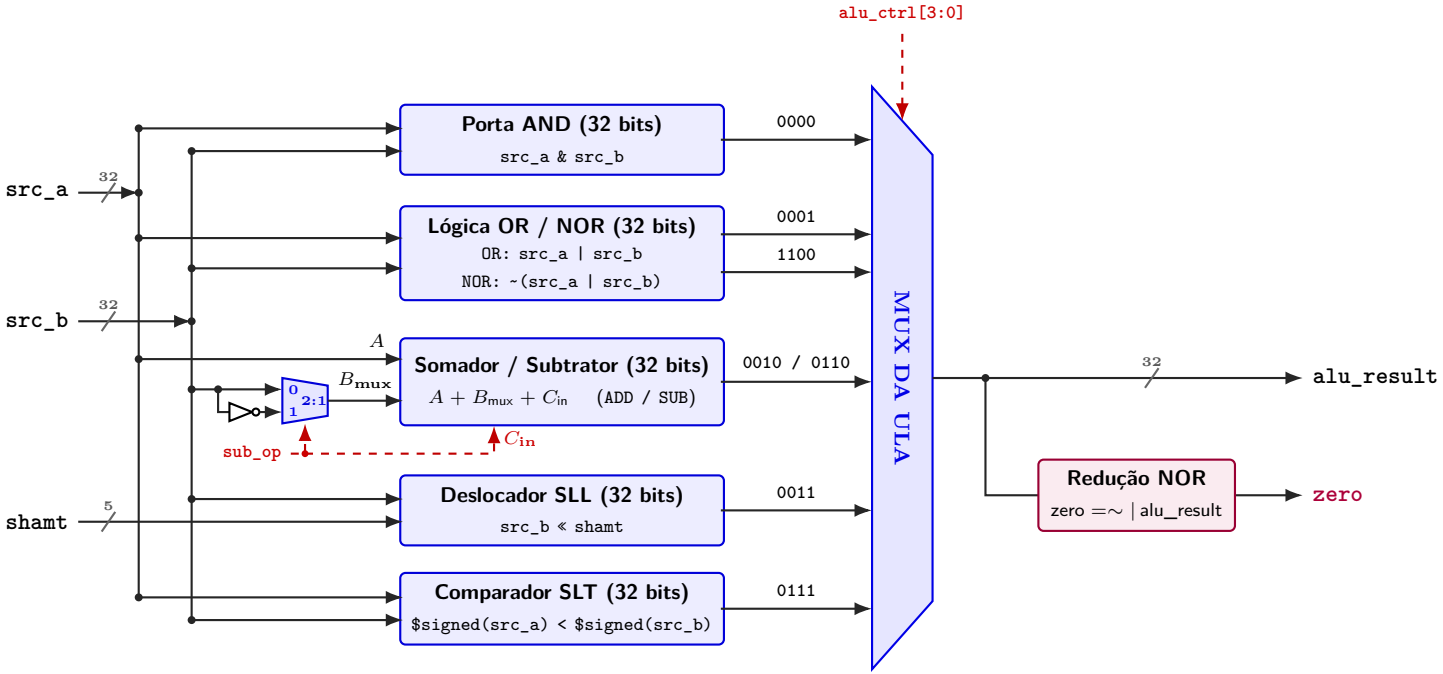
alu_ctrl[3:0]
src_a
32
src_b
32
shamt
5
Porta AND (32 bits)
src_a & src_b
0000
Lógica OR / NOR (32 bits)
OR: src_a | src_b
NOR: ~(src_a | src_b)
0001
1100
A
B_{mux}
0
1
2:1
sub_op
C_{in}
Somador / Subtrator (32 bits)
$A + B_{mux} + C_{in}$ (ADD / SUB)
0010 / 0110
Deslocador SLL (32 bits)
src_b ≪ shamt
0011
Comparador SLT (32 bits)
$signed(src_a) < $signed(src_b)
0111
MUX DA ULA
32
alu_result
Redução NOR
zero =∼ | alu_result
zero